

An AI that studies you, so you can study smarter.

ParikshaMitra (परीक्षा मित्र)

An Agentic AI Study Companion for Indian Competitive Exams

The Mission

Autonomously plan, test, analyze, and adapt student preparation.

The System

A multi-agent cognitive architecture built on LangGraph and Gemini.

The Origin

Built by Team ParikshaMitra for Reskill x JIT Borawan Hackathon.

India's Exam Preparation Crisis

40+ Million
students compete
annually.

Example: 24 Lakh NEET applicants fight for just 1.25 Lakh seats (a 20:1 ratio).



One-Size-Fits-All

Static 4-7 hour coaching schedules ignore individual pacing, strengths, and weaknesses.



No Diagnostic Feedback

Mock tests output basic scores and percentiles, failing to explain why a student failed or what to do next.

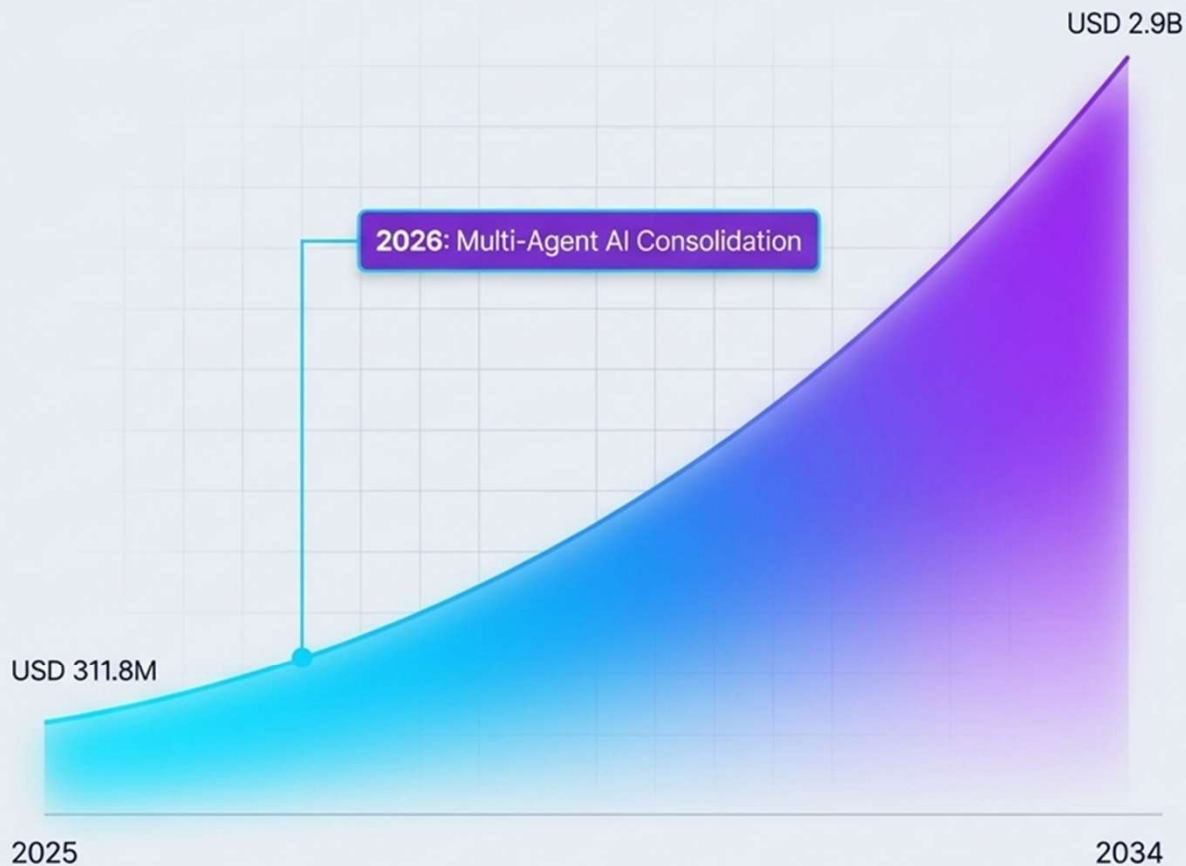


Information Overload

Students drown in endless PDFs and YouTube videos without an orchestrator to sequence their learning.

The Core Insight: The problem isn't a lack of educational content; it's the absence of an intelligent system to dynamically orchestrate learning.

The \$3 Billion Gap in EdTech



Massive Growth

India's test prep market surges from USD 311.8M (2025) to USD 2.9B (2034) at a 26.8% CAGR.

The Competitive Gap

Incumbents (PW, Unacademy, Allen) rely entirely on passive lectures and static test series.

The Timing (Why Now)

2026 is the consolidation year for multi-agent systems, shifting EdTech from human-dependent doubt resolution to autonomous orchestration.

First-Mover Advantage

ParikshaMitra is positioned as the first closed-loop system offering autonomous planning and adaptive testing.

True Agentic Autonomy



4+1 Cognitive Architecture



The Intelligence Engine



Spaced Repetition (SM-2)

Dynamically calculates optimal review days.

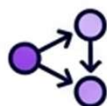
>80% score: 2x interval. 50-80%: maintain.
<50%: reset to 1 day.



Adaptive Difficulty (CAT-lite)

Adjusts question difficulty (Level 1-5) in real-time.

Moves up 1 level for 3 consecutive correct answers;
drops 1 level for 2 wrong.



Concept Dependency DAG

Checks prerequisite mastery before advancing.

E.g., System verifies 'Vectors' mastery if a student fails
'Newton's Laws'.



Exam Readiness Score

A predictive 0-100 metric for quantifiable confidence.

$(0.4 \times \text{Coverage}) + (0.3 \times \text{Mastery}) + (0.2 \times \text{Revision}) +$
 $(0.1 \times \text{Mock Trend}).$

Engineered for Speed & Scale

Built with **GitHub Copilot**

Agent Framework & Logic

LangGraph

Powers native stateful workflows, conditional routing, and human-in-the-loop checkpoints.

Python 3.11

Core backend logic.

LLM Core

Google Gemini 2.0 Flash

High-speed structured JSON outputs for quizzes and native multilingual support.

App Layer & State

Streamlit

Rapid UI prototyping and built-in visual charting.

SQLite + Pydantic v2

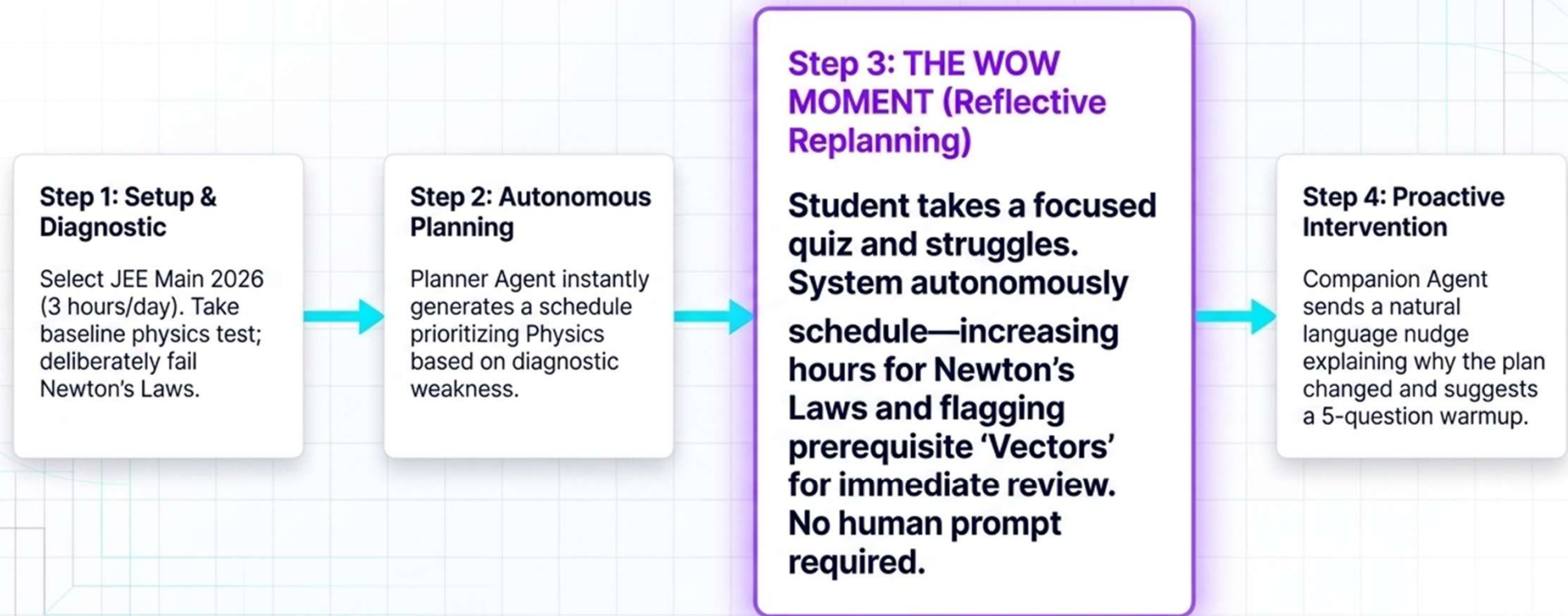
Persistent student history and type-safe data modeling.

GitHub Copilot used extensively for quiz prompt scaffolding, API endpoint generation, and UI layout.

Hackathon MVP Capabilities



The ParikshaMitra Experience



Beyond Static Chatbots

	Traditional Platforms	Static AI	ParikshaMitra
Study Planning	Generic schedules	Responds if asked	Autonomous adaptive planning
Quiz Difficulty	Fixed levels	Same for all	Real-time adaptation (CAT-lite)
Feedback	Just scores	Explains if asked	Proactive prerequisite diagnosis
Adaptation	None	Per-conversation only	Persistent cross-session knowledge state
Agentic Score	0/6	1/6	6/6 Agentic Properties

Democratizing Elite Preparation



Educational Impact

Higher Outcomes

Targeted adaptive practice yields a 30-40% improvement in weak topics.

Time Efficiency

Cuts wasted time by 50% through intelligent SM-2 spaced repetition scheduling.



Social Impact

Accessible Excellence

A student in a Tier-3 city gets the same hyper-personalized orchestration as a premium Kota student.

Radical Affordability

API costs are < Rs 1 per student per day, making elite tutoring virtually free.

NEP 2020 Aligned

Directly supports India's push for personalized, competency-based education.

The Vision Forward

Phase 2 (Months 2-3)

Expand to NEET/UPSC, add time-management analytics and peer comparison.

Phase 3 (Months 4-6)

Voice interaction, WhatsApp bot, and regional languages (Marathi, Tamil).

Phase 4 (Months 6-12)

B2B integration plugins for major coaching centers; parent dashboards.

Revenue Model

Freemium: Free core app to rapidly build the user base.

Premium: Rs 99/month for unlimited adaptive quizzes and full SM-2 analytics.

B2B: Institutional licensing for schools and coaching centers.